

MAMADOU SAIDOU BALDE

AI Engineer | Data Scientist

Paris, France • +33 7 53 26 68 07 • baldeba772@gmail.com • [LinkedIn](#) • [GitHub](#) • [Hugging Face](#)

PROFESSIONAL SUMMARY

AI Engineer & Data Scientist with 5+ years of end-to-end experience designing and deploying production-grade AI solutions. Specialized in NLP, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), AI Agents, and MLOps. Proven track record in demanding industrial environments (Nokia 5G), delivering scalable, robust systems through complete CI/CD pipelines: from raw data exploration to cloud deployment. Business-value driven with a strong focus on performance and reliability.

TECHNICAL SKILLS

AI / ML / NLP	LLMs, RAG, AI Agents, MCP, NLP, Computer Vision, Fine-tuning, Prompt Engineering, LangChain
Frameworks	PyTorch, TensorFlow, Scikit-learn, Transformers (BERT, GPT, Mistral), Hugging Face, FAISS
Languages	Python, SQL, PySpark, Scala, R, SAS, C
Data & BI	Power BI, Tableau, Looker Studio, PostgreSQL, MySQL, MongoDB, Neo4j, MSSQL (SSAS/SSIS/SSRS)
Big Data	Apache Spark, Hadoop, HDFS, Databricks
Cloud / MLOps	AWS (SageMaker, EC2, S3, Lambda, ECS), Azure ML, GCP (BigQuery, Vertex AI)
DevOps / Tools	Docker, Kubernetes, MLflow, Jenkins, Git/GitHub, Gerrit, CI/CD
Methodology	Agile (Scrum), Jira, Confluence

PROFESSIONAL EXPERIENCE

AI Engineer | Data Scientist | Nokia Corporation - Massy-Palaiseau, France

September 2021 - April 2025

Member of the CI_Metrics Squad within the Mobile Networks division, improving stability and performance of the CI/CD (Continuous Integration) pipelines for 5G L2/L3 network deployments in a global Agile and international environment.

- ▶ Extracted and processed very large volumes of complex data from diverse sources to monitor and improve software build stability.
- ▶ Developed custom KPI metrics to evaluate system performance (compilation time, success rates, pipeline stability).
- ▶ Built Machine Learning models to predict build failures, reduce errors, and anticipate pipeline outages.
- ▶ Designed interactive Power BI dashboards to track key operational indicators (build time, success rate) for stakeholders.
- ▶ Implemented robust ETL (Extract - Transform - Load) pipelines to automate real-time data extraction, transformation, and loading using Python, SQL, Spark, and PowerBI.
- ▶ Led the design and deployment of an automated system tracking code evolution (lines added, removed, modified) across repositories.

Data Scientist | Analyst | Michelin - Clermont-Ferrand, France

February 2021 - June 2021

R&D project analysing driving behaviour from embedded telematics data to optimise vehicle recognition.

- ▶ Analysed a dataset of 2.7M observations (28 variables, 40 vehicles) from sensors capturing 3-axis acceleration, speed, weather, road type, and time of day.
- ▶ Performed feature engineering to extract behavioural indicators (aggressiveness, stability, longitudinal/lateral variations).
- ▶ Developed Machine Learning models to optimise vehicle recognition accuracy.
- ▶ Identified behavioural patterns linked to weather conditions and trip schedules.

Data Scientist | Analyst | Blaise Pascal Mathematics Laboratory - Aubière, France

April 2020 – August 2020

Statistical study of the socio-economic determinants of crime rates in France using departmental data.

- ▶ Conducted exploratory analysis and statistical modelling across 96 French departments (correlations, clustering, multivariate analysis).
- ▶ Studied rates of crimes against persons, burglaries, and car thefts in relation to socio-economic variables (unemployment, poverty, urbanisation, single-parent households).
- ▶ Identified significant correlations between crime levels, economic precarity, and urban density.
- ▶ Produced analytical reports and visualisations for laboratory researchers and partner authorities.

FREELANCE PROJECT - AI ENGINEER

Clinical-Safe Medical AI Chatbot - NLP / LLM / RAG - Deployed on Hugging Face Spaces

End-to-end RAG system built for a medical PhD student to securely query confidential research documents - without any external data leakage.

Context & Challenge

- ▶ Identified a real-world need: a medical PhD student struggled to navigate large volumes of confidential research PDFs efficiently.
- ▶ Key constraints: strict data privacy (no external API calls), answer reliability (no hallucination), and clinical tone accuracy.

Architecture & Technical Implementation

- ▶ Designed a full RAG pipeline: PDF ingestion → chunking (RecursiveCharacterTextSplitter) → BAAI/bge-small-en embeddings → FAISS vector store → semantic retrieval → LLM generation.
- ▶ Integrated Qwen2.5-0.5B-Instruct via HuggingFacePipeline with a custom clinical-safe prompt enforcing strict grounding: the model answers only from retrieved context.
- ▶ Implemented a similarity score threshold (FAISS) to detect out-of-scope questions and return an explicit fallback message instead of hallucinated answers.
- ▶ Added ConversationBufferWindowMemory (k=3) for natural multi-turn dialogue while preventing context drift.
- ▶ Built post-processing logic to clean and deduplicate LLM outputs, removing prompt leakage and repetition artefacts.

Evaluation & Quality Assurance

- ▶ Implemented a RAGAS evaluation pipeline (faithfulness, answer relevancy, context relevancy) to measure RAG quality across a test question set.
- ▶ Iterated on chunk size, overlap, score threshold, and prompt design based on evaluation results.

Deployment

- ▶ Deployed on Hugging Face Spaces with a Gradio UI; optimised for constrained CPU environments (thread management, float32 fallback, HF cache redirection to /tmp).

Stack: Python · LangChain · Transformers · FAISS · Qwen2.5 · BAAI/bge · Gradio · HuggingFace Spaces · RAGAS

EDUCATION

Master's Degree - Data Science & Artificial Intelligence | Data ScienceTech Institute (DSTI) - Nice, France

September 2021 – November 2023

Master's Degree - Applied Mathematics & Statistics | Université Clermont Auvergne - Clermont-Ferrand, France

September 2019 – April 2021

Bachelor's Degree - Mathematics | Aix-Marseille Université - Marseille, France

September 2018 – April 2019

LANGUAGES & INTERESTS

English: Fluent - Cambridge Certification

French: Native

Interests: Strength training, marathon running, football (active team member) · Reading (data science, AI, self-development) · Music, manga